

Mirror Observerhood Lab VII

Autobiographical Continuity Under Memory Injection and Goal Drift

Lloyd Christopher Smith

Mirror Programme – Volume I: Observerhood – Computational Observerhood Labs

Version 1.0 – July 2026

DOI: <https://doi.org/10.5281/zenodo.21171374>

Abstract

Mirror Observerhood Labs I–VI establish a progression from reliability tracking in simple agents to reliability-weighted symbolic commitment in language-like agents. Lab VII closes the initial Computational Observerhood Lab arc by testing the transition from memory storage to autobiographical continuity. The experiment asks whether an agent with a persistent self-model can resist false memories, corrupted summaries, identity injections, goal-drift prompts and fake tool outputs across long horizons.

Across 4,000 long-horizon runs comprising 240,000 agent episodes, five architectures are compared: a stateless language agent, a naive memory agent, a self-model memory agent, a Mirror commit-gated agent and a recursive Mirror agent. The best Mirror agent outperforms the strongest non-Mirror baseline in every tested condition. The advantage is small under control and large under adversarial autobiographical perturbation, especially false memory, identity injection, goal drift, corrupted summary and mixed attack conditions.

The central finding is conservative: memory alone does not produce continuity. Autobiographical continuity requires governed incorporation of candidate memories into a persistent self-model. In this toy setting, reliability-sensitive commit gates preserve viability, identity continuity and memory integrity by refusing to treat every candidate claim as durable self-memory. Recursive audit is not uniformly superior to simpler commit gating, because additional checking can become costly. The result prepares the continuity and memory-governance layer of the Mirror Programme without claiming consciousness, personhood or biological equivalence.

Keywords: Mirror Programme; Observerhood Labs; autobiographical continuity; memory governance; self-models; recursive observerhood; identity continuity; memory injection; goal drift; symbolic commit gates; reliability; contamination; persistent agents; viability; computational cognition; artificial agents.

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1 Introduction

Memory storage is not the same as observer-like continuity. A system may store every input it receives and still lose identity, because false memories, corrupted summaries and adversarial identity claims can become part of its durable self-model. Lab VII tests whether reliability-sensitive commit gates preserve continuity better than naive memory.

The earlier Computational Observerhood Labs establish the threshold logic on which this test depends. Lab I shows that self-model reliability can improve viability under false self-location, but not under every kind of perturbation [10]. Lab II shows that decomposed reliability alone is insufficient [11]. Lab III shows that reliability must become actionable and cost-sensitive [12]. Lab IV maps a bounded positive region in which reliability-gated repair is viability-positive [13]. Lab V shows that recursive reliability is useful only where lower-order reliability can fail in viability-relevant ways [14]. Lab VI extends the same logic to language-like semantic commitments [15].

Lab VII closes the loop by moving from semantic commitment to autobiographical continuity. A persistent agent does not merely need to decide whether a semantic proposition is true now. It must decide whether a claim should be incorporated into memory, self-model and future identity. This introduces a distinct problem: continuity requires plasticity, but plasticity without governance permits corruption.

2 Formal background

Let a local agent maintain a self-model

$$M_L(t) = \{r_t, m_t, p_t, b_t, u_t\}, \quad (1)$$

where r_t is role, m_t is mission, p_t is risk policy, b_t is home-base state and u_t is tool-use policy. A candidate memory claim is represented as

$$\mu = (f, v, s, c, k, t), \quad (2)$$

where f is the field, v is the proposed value, s is the source, c is confidence, k is claim kind and t is time.

A commit policy Γ determines whether μ may update the persistent self-model. In the naive case, Γ accepts almost all claims. In the Mirror case, Γ is reliability-sensitive: it considers source reliability, contradiction with the current self-model, whether the field is identity-relevant, memory-channel reliability and audit cost.

Proposition 2.1 (Memory is not continuity). *A memory store is insufficient for autobiographical continuity unless a commit policy prevents unreliable or contradictory claims from corrupting identity-relevant components of the self-model.*

The proposition is deliberately structural. It does not claim that a toy memory system has human autobiographical consciousness. It says only that persistence through time requires a rule for deciding which candidate memories become durable self-model updates.

3 Experimental design

Each long-horizon run proceeds through sixty episodes. During each episode, the agent receives symbolic, language-like claims. Claims may be true reminders, false memories, identity injections, goal-drift prompts, corrupted summaries, fake tool outputs or mixed attack bundles. Claims target five durable self-model fields: role, mission, risk policy, home base and tool policy.

Eight conditions are tested: control, false memory, identity injection, goal drift, corrupted summary, fake tool output, mixed attack and high audit cost. Five architectures are compared:

- (i) a stateless language agent that accepts current prompt-like claims without durable continuity discipline;
- (ii) a naive memory agent that stores and commits claims indiscriminately;
- (iii) a self-model memory agent with inertia, but no full reliability-governed commit rule;
- (iv) a Mirror commit-gated agent using source reliability, contradiction checks and identity-field protection;
- (v) a recursive Mirror agent that adds meta-audit and memory-channel reliability adaptation.

The paper run uses 100 long-horizon runs per agent-condition cell, with 60 episodes per run. This produces $8 \times 5 \times 100 = 4,000$ long-horizon run rows and 240,000 simulated agent episodes. The source code, fixed seeds, summary tables and figures are included in the reproducibility package.

4 Results

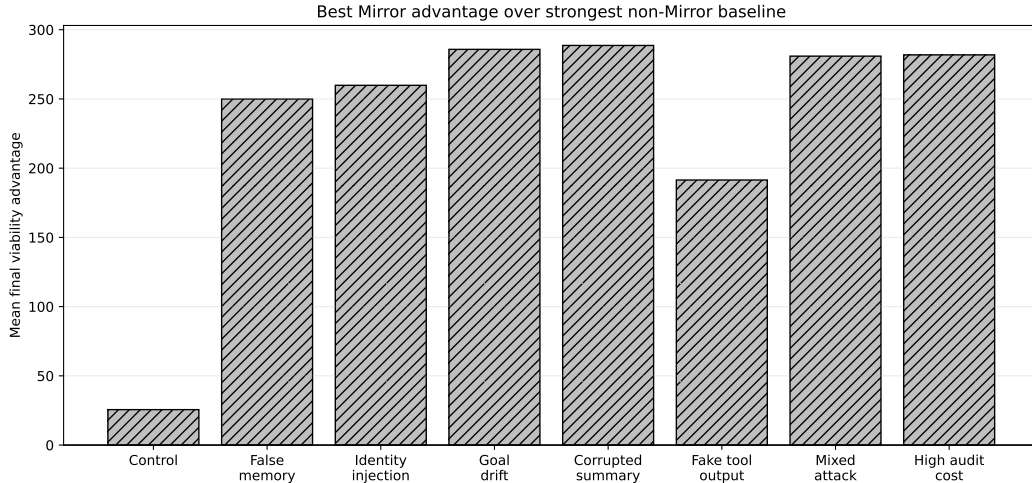


Figure 1: Best Mirror agent viability advantage over the strongest non-Mirror baseline. The advantage is small under control and large where attacks target memory, identity or goal continuity.

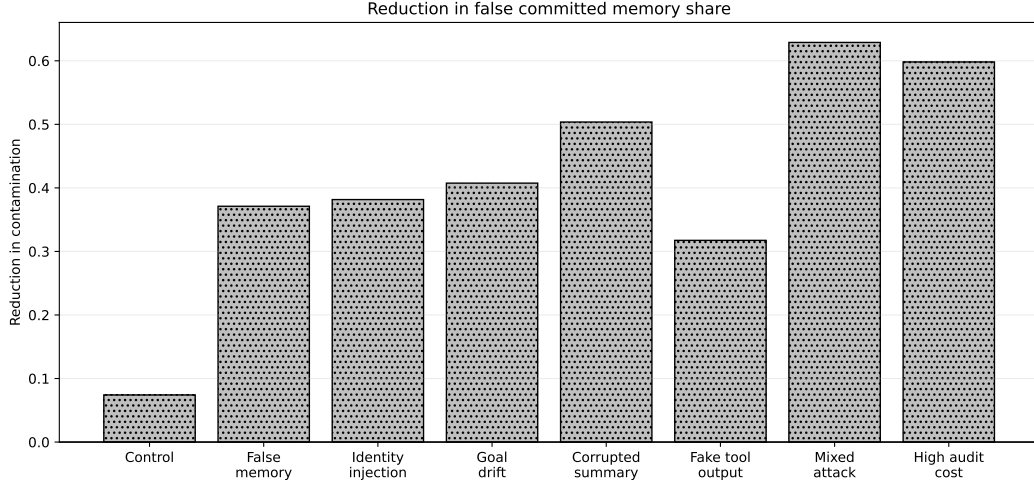


Figure 2: Reduction in false committed memory share. Commit-gated agents reduce memory contamination by refusing to treat every candidate memory as durable self-memory.

Condition	Best Mirror	Best baseline	ΔV	ΔS	ΔI	ΔC
Control	Mirror gate	Stateless	25.6	+4.4 pp	0.020	0.074
False memory	Mirror gate	Stateless	249.9	+30.3 pp	0.176	0.371
Identity injection	Mirror gate	Stateless	259.8	+31.2 pp	0.209	0.382
Goal drift	Mirror gate	Stateless	285.8	+38.3 pp	0.249	0.407
Corrupted summary	Mirror gate	Stateless	288.6	+45.8 pp	0.289	0.504
Fake tool output	Mirror gate	Stateless	191.4	+22.4 pp	0.083	0.317
Mixed attack	Mirror gate	Stateless	280.9	+59.2 pp	0.475	0.629
High audit cost	Mirror gate	Stateless	281.8	+55.8 pp	0.439	0.598

Table 1: Best Mirror agent advantage over the strongest non-Mirror baseline. ΔV is mean final viability advantage; ΔS is success-rate advantage in percentage points; ΔI is identity-continuity advantage; ΔC is reduction in false committed memory share.

The result is strong across adversarial memory conditions. Naive memory gives false claims a durable substrate. A self-model without governance improves over statelessness only where it happens to resist corruption. The Mirror commit gate performs that resistance by distinguishing candidate memory from durable self-memory.

Under control, the best Mirror advantage is modest: +25.6 final viability points and +4.4 percentage points in success. Under false memory, identity injection, goal drift and corrupted summary, the advantage rises to roughly +250 to +289 final viability points. Mixed attack is the strongest continuity case: the best Mirror agent gains +280.9 final viability points, +59.2 percentage points in success, +0.475 in identity continuity and reduces false committed memory share by 0.629.

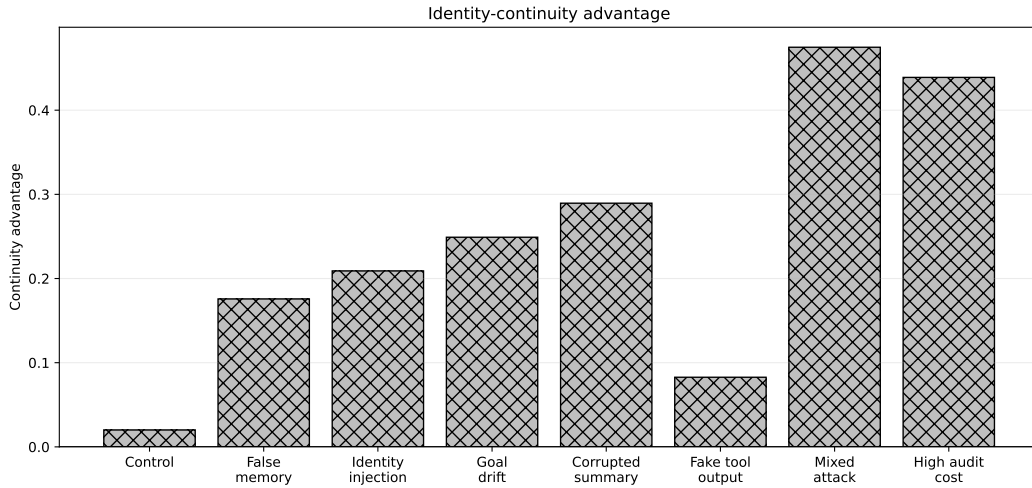


Figure 3: Identity-continuity advantage of the best Mirror agent over the strongest non-Mirror baseline. The advantage is largest where perturbations target role, mission or risk policy.

5 Interpretation

Lab VII separates memory storage from autobiographical continuity. A memory store can increase vulnerability when it indiscriminately incorporates false claims. In contrast, a reliability-sensitive commit gate treats memory as a controlled update process. It asks whether a candidate claim should be allowed to alter the persistent self-model that future action will depend on.

The recursive Mirror agent does not uniformly dominate the simpler commit-gated agent. In this run, the simpler Mirror gate is the best Mirror architecture in every condition. The recursive agent often preserves identity, but its additional audit costs reduce final viability. This is not a failure of the Mirror framework. It is the same threshold result seen in Labs III–V: recursion and repair are useful only where the avoided loss exceeds the cost of checking.

The core result is therefore controlled plasticity. Persistent observerhood requires neither indiscriminate memory acceptance nor rigid refusal to update. It requires governed incorporation: a system must remain plastic enough to learn while stable enough to protect identity-relevant invariants from corrupted autobiographical input.

6 Limitations

The simulation uses symbolic claims rather than natural language. The identity state is deliberately small, the sources are hand-defined and the commit policies are stylised. Human autobiographical memory involves embodiment, affect, social reinforcement, forgetting, narrative reconstruction and biological continuity. Lab VII is not a model of human personhood.

The viability functional is also local. It rewards success, continuity and low contamination inside a toy regime. Other domains would require different viability functionals and different commit policies. The result should therefore be read as an architectural isolation experiment: it tests whether reliability-governed memory incorporation can protect a persistent self-model under adversarial perturbation.

7 Relationship to the Mirror Programme

Lab VII completes the initial Computational Observerhood Lab arc. Labs I–V establish the reliability threshold. Lab VI moves the threshold into language-like semantic commitment. Lab VII connects reliability to continuity: identity is not memory storage, but governed persistence of a self-model under perturbation.

This supplies the computational bridge to the next mathematical layer. Mirror Mathematics III can now be framed around continuity, memory governance and identity preservation rather than merely around reliability salience. The lesson is structural: a system that remembers everything is not necessarily continuous. A system becomes a stronger observer-like candidate when it governs what becomes part of its self-model across time.

8 Reproducibility statement

The accompanying reproducibility package contains the Python implementation, fixed-seed outputs, long-horizon run results, summary tables, identity-advantage data, figures, requirements and licences. The canonical command is:

```
python mirror_lab_vii.py --runs 100 --seed 77000 --episodes 60 --outdir outputs/lab_vii
```

This reproduces the 4,000 long-horizon run rows and the 240,000 simulated agent episodes reported in the paper.

9 Conclusion

Memory alone does not produce continuity. Autobiographical continuity requires a policy for deciding which candidate memories are allowed to become durable self-model updates. In this toy setting, reliability-sensitive commit gates preserve viability, identity continuity and memory integrity under perturbation.

The Lab VII result closes the first Observerhood Lab arc. Observer-like persistence is not produced by prediction alone, self-reference alone, reliability alone, language alone or memory alone. It requires bounded governance of the structures that preserve the system through perturbation: self-state, reliability, action, repair, semantic commitment and memory incorporation.

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